GCSE GEOGRAPHY

Paper 2: Challenges in the human environment

Topic 6: Resource management

Practice questions in the style of AQA GCSE Geography (8035). Every question is original; the layout, command words, mark allocations (including levels-based 6- and 9-mark questions with SPaG) and mark scheme follow the real exam format. Figures and data are illustrative and simplified for practice; case-study facts in the mark scheme are summarised, so check them against your own case-study notes.

Time allowed	About 100 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a black pen, a pencil, a ruler, a calculator and (for some questions) a protractor.
Instructions	• Use black ink or black ball-point pen. Use pencil only for drawing. • Answer **all** questions. • You must answer the questions in the spaces provided. • Do all rough work in this book. Cross through any work you do not want to be marked. • Questions that carry a **SPaG** note are also marked for spelling, punctuation, grammar and the use of specialist terminology.
Information	• The maximum mark for this paper is **95**. • The marks for questions are shown in brackets. • Questions that start with the command word *Explain* need developed points (a point plus a reason or consequence). • 6- and 9-mark questions are marked by **levels**: aim to give a detailed, balanced and well-linked answer, using place-specific detail.

Name: _____ Date: _____ Mark: _____ / 95

Section C: Resource management

Answer all questions in this section.

0 1 This question is about food, water and energy. **Figure 1** shows the average daily calorie supply per person in five regions.

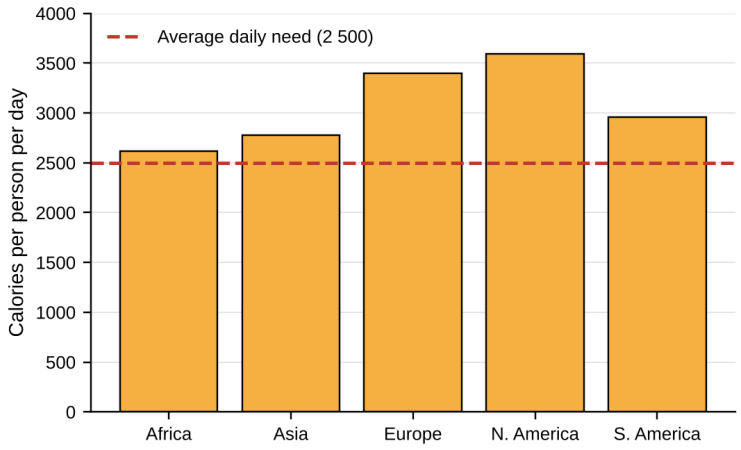
0 1 . 1 What is meant by **resource management**? **[1 mark]**

*Tick **one** box.*

- ☐ adding more resources to the environment
- ☐ managing the supply and use of resources
- ☐ stopping all use of natural resources
- ☐ making resources cheaper

0 1 . 2 Which region in Figure 1 has the highest daily calorie supply? **[1 mark]**

Figure 1



0 1 . 3 Calculate how many more calories per day are available to a person in Europe than to a person in Africa. **[2 marks]**

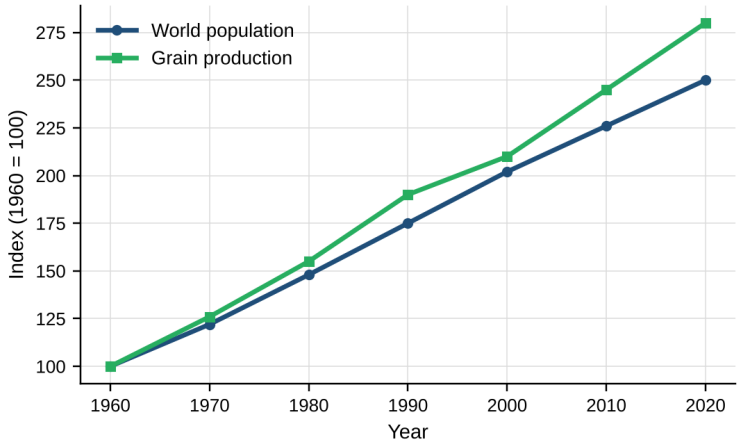
Answer = _____ calories

0 1 . 4 Describe how the food supply in the world varies between regions. **[3 marks]**

0 1 . 5 Explain **two** causes of food insecurity. **[4 marks]**

0 1 . 6 **Figure 2** shows world population and grain production. Describe what the figure shows and suggest why grain production has grown faster than the population. **[4 marks]**

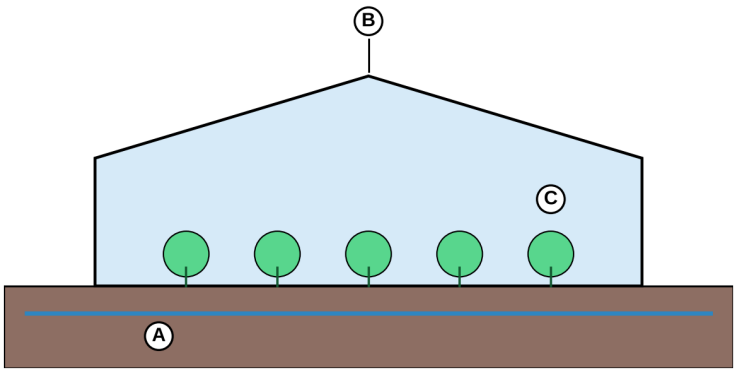
Figure 2



0 1 . 7 Describe **two** impacts of food insecurity on people. **[4 marks]**

0 1 . 8 **Figure 3** shows a greenhouse used for large-scale food production. Name feature **A** and explain how it helps to increase food supply. **[3 marks]**

Figure 3



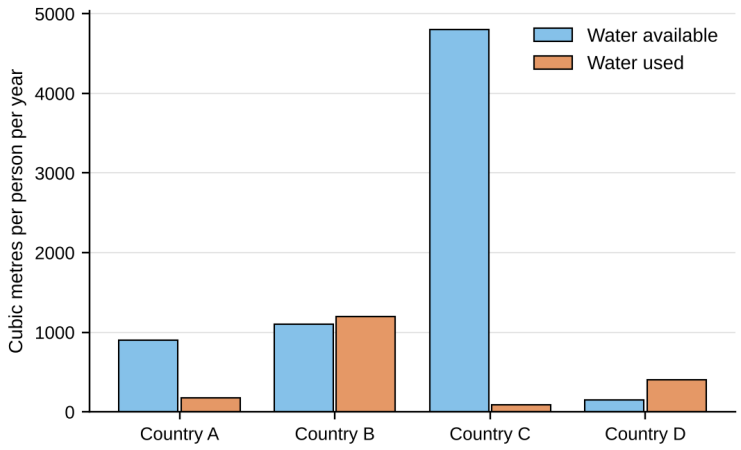
0 1 . 9 Explain how **one** strategy could increase food supply (e.g. irrigation, aeroponics or hydroponics, new technology, or the use of biotechnology). **[3 marks]**

0 1 . 10 Evaluate the use of **agribusiness** or **large-scale agricultural schemes** in increasing food supply. Refer to a named example. **[6 marks]**

0 2 This question is about water. **Figure 4** shows water availability and use in four countries.

0 2 . 1 Which country in Figure 4 has the greatest water surplus? Give the figure. **[2 marks]**

Figure 4



0 2 . 2 Which **two** countries in Figure 4 have a water deficit? **[2 marks]**

0 2 . 3 Explain **two** reasons why the demand for water has increased in many countries. **[4 marks]**

0 2 . 4 Describe **two** ways in which water supply can be increased. **[4 marks]**

0 2 . 5 Explain how **over-abstraction** of groundwater causes problems. **[3 marks]**

0 2 . 6 Describe the main features of a **large-scale water transfer scheme** you have studied (e.g. the Lesotho Highlands Water Project). **[4 marks]**

0 2 . 7 Explain the advantages and disadvantages of a large-scale water transfer scheme you have studied. **[6 marks]**

0 2 . 8 **'Managing water supplies sustainably is more important than increasing supply.'** To what extent do you agree?

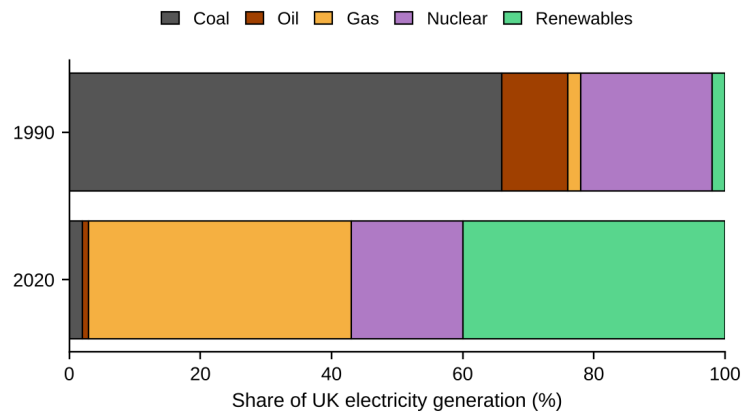
[9 marks]
+ [3 marks]
SPaG

Spelling, punctuation, grammar and use of specialist terminology [3 marks]

0 3 This question is about energy. **Figure 5** shows how the UK's electricity generation changed between 1990 and 2020.

0 3 . 1 State the percentage of UK electricity that came from renewables in 2020. **[1 mark]**

Figure 5

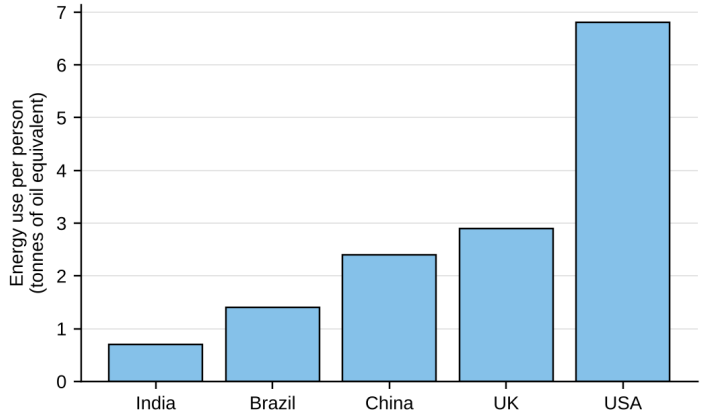


0 3 . 2 Describe **two** changes in the UK's energy mix between 1990 and 2020. **[2 marks]**

0 3 . 3 Explain **two** reasons for the change in the UK's energy mix. **[4 marks]**

0 3 . 4 **Figure 6** shows energy use per person in five countries. Suggest **two** reasons why energy use per person is higher in the USA than in India. **[2 marks]**

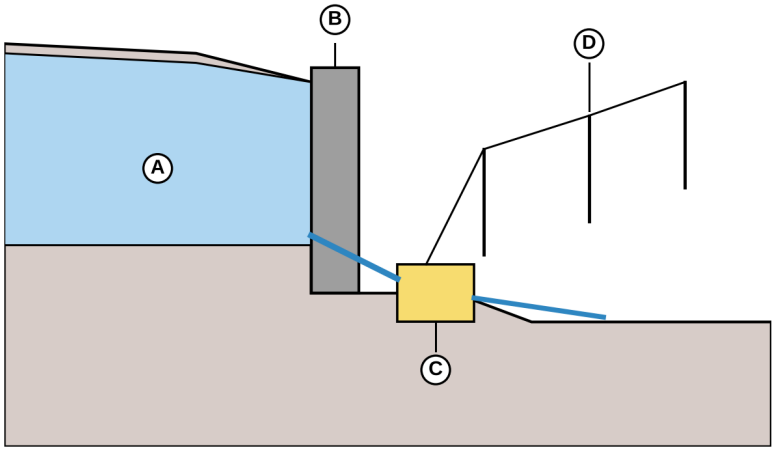
Figure 6



0 3 . 5 Explain how **one** fossil fuel and **one** renewable source of energy can have environmental impacts. **[4 marks]**

0 3 . 6 **Figure 7** shows a hydro-electric power (HEP) scheme. Name features **A** and **B** and explain how electricity is generated. **[4 marks]**

Figure 7



0 3 . 7 Describe **two** ways in which individuals or groups can reduce energy consumption. **[4 marks]**

0 3 . 8 Explain how **energy conservation** and **increasing renewable energy** could help a country meet its energy needs sustainably. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 6: Resource management

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *geography*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	controlling how resources are used / supply and demand of resources	Tick box 2 is correct.	1	AO1 3.2.3.1
01.2	North America		1	AO3 3.2.3.1
01.3	3 400 − 2 620 ; 780 (calories) ;	Allow 750–810.	2	AO4 3.2.3.1
01.4	Africa has the lowest supply (about 2 620) / N. America highest (about 3 590) ; Europe and North America are well above the daily need ; all regions are above the 2 500-calorie average need but distribution is uneven within regions / some people still go hungry ;		3	AO3 3.2.3.2
01.5	climate change / droughts / floods reduce crop yields ; conflict disrupts farming and transport ; poverty: people cannot afford to buy food ; pests and diseases destroy crops ; population growth increases demand ; soil degradation / lack of water ;	Two causes, each with 1 mark for the cause and 1 for the development.	4	AO2 3.2.3.2
01.6	grain production rose from 100 to 280 (1960–2020), population from 100 to 250 ; so grain production has grown faster than the population ; reasons: Green Revolution / high-yield varieties / fertilisers / irrigation / mechanisation ; more land used for farming ;		4	AO2/AO3 3.2.3.2
01.7	famine / starvation / malnutrition, causing deaths ; undernutrition: stunted growth, weakened immune system, so disease spreads ; high food prices cause social unrest / riots ; soil erosion as farmers clear land ; people leave their homes (migration) ;		4	AO1/AO2 3.2.3.2
01.8	A: irrigation system / drip irrigation / water pipe ; supplies water directly to the roots, so little is lost ; crops can grow in dry areas / all year, so yields increase ;	B = glass / plastic roof (keeps in heat), C = crops.	3	AO2 3.2.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.9	irrigation brings water to dry areas, allowing farming in arid places ; hydroponics grows plants in nutrient solution so does not need soil ; GM crops with higher yields / pest-resistant ; so more crops are grown per area, and more food is produced ;		3	AO2 3.2.3.3
01.10	**Level 3 (5–6 marks)**: Detailed, accurate and relevant knowledge and understanding of the topic. Clear, well-developed and logically structured explanation/analysis with specific links made between ideas (and a use of place detail or the resources where asked). **Level 2 (3–4 marks)**: Some relevant and accurate knowledge and understanding. Some explanation/analysis, but developed unevenly or with limited links. Some use of detail or the resources. **Level 1 (1–2 marks)**: Basic, general knowledge with limited accuracy or relevance. Simple statements; little or no explanation or use of detail. **0 marks**: No relevant content.		6	AO1/AO3 3.2.3.3
	Indicative content (credit any other valid geography, and place-specific detail): • named example e.g. Almería, Spain (greenhouses, 'sea of plastic') or Indian Green Revolution ; • benefits: high yields all year round / jobs / income / exports ; • uses technology: drip irrigation, hydroponic, fertilisers ; • problems: water shortage / groundwater depleted ; plastic waste ; pollution from chemicals ; • workers often low paid / migrant workers ; • judgement: increases food supply but needs sustainable management ;			
02.1	Country C ; 4 710 (m^3 per person per year) allow 4 700–4 720 ;	Surplus = available − used: A 720, B −100, C 4 710, D −250.	2	AO3 3.2.3.4
02.2	Country B ; Country D ;	Use > available.	2	AO3 3.2.3.4
02.3	population growth: more people need water for drinking / sanitation ; rising living standards: more showers, washing machines, swimming pools ; industrialisation: factories need water ; irrigation for agriculture ; more meat in diets requires more water ;		4	AO2 3.2.3.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	building dams and reservoirs to store water ; water transfer schemes: pipelines / tunnels move water from surplus to deficit areas ; desalination: removing salt from seawater ; groundwater abstraction (wells / boreholes) ; rainwater harvesting ;		4	AO1 3.2.3.5
02.5	water is taken faster than it is replaced, so the water table falls ; wells and rivers dry up / wetlands lost ; near coasts salt water can enter aquifers / ground may subside ;		3	AO2 3.2.3.5
02.6	named scheme / location e.g. Lesotho Highlands: dams (Katse, Mohale) and tunnels in Lesotho that transfer water to South Africa (Gauteng) ; purpose: provides water for Johannesburg / industry and HEP for Lesotho ; scale e.g. 200 km of tunnels and several dams ; partners / funding ;		4	AO1 3.2.3.5
02.7	**Level 3 (5–6 marks)**: Detailed, accurate and relevant knowledge and understanding of the topic. Clear, well-developed and logically structured explanation/analysis with specific links made between ideas (and a use of place detail or the resources where asked). **Level 2 (3–4 marks)**: Some relevant and accurate knowledge and understanding. Some explanation/analysis, but developed unevenly or with limited links. Some use of detail or the resources. **Level 1 (1–2 marks)**: Basic, general knowledge with limited accuracy or relevance. Simple statements; little or no explanation or use of detail. **0 marks**: No relevant content.		6	AO1/AO3 3.2.3.5
	Indicative content (credit any other valid geography, and place-specific detail): • advantages: clean water for millions in the receiving area ; jobs during construction / income (royalties) to Lesotho ; HEP generates electricity ; • improved roads / services in remote area ; • disadvantages: displacement of villages / loss of farmland and grazing ; • environmental damage: loss of habitats / downstream river flow reduced ; • very costly / debt ; benefits mostly go to South Africa ; • judgement on whether the benefits outweigh costs ;			

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.8	**Level 3 (7–9 marks)**: Detailed, accurate knowledge and understanding of the topic with a well-developed, balanced and logical argument. A clear, supported **evaluation/judgement** that is based on evidence (and specific place detail or the resources where asked). Any conclusion is fully justified. **Level 2 (4–6 marks)**: Some accurate knowledge and understanding. Some development of ideas with evidence; the judgement/evaluation is partial, unbalanced or not fully supported. **Level 1 (1–3 marks)**: Limited knowledge and understanding. Simple statements with little development; a judgement, if any, is unsupported. **0 marks**: No relevant content.		9 + 3 SPaG	AO1/AO2/AO3 3.2.3.5
	Indicative content (credit any other valid geography, and place-specific detail): • Agree: sustainable management means water is available in the future ; examples: rainwater harvesting, recycling water, water meters ; • Agree: groundwater abstraction and dams harm environments and are costly ; reduces conflict between countries ; • Disagree: in many places demand exceeds supply, so increase in supply is needed (dams, transfers, desalination) ; • Disagree/balance: poorer countries cannot afford the technology and have a growing population – they need new supply ; • Evaluation: a mix of increasing supply and sustainable management is best ; reasoned judgement with named example (e.g. Lesotho Highlands, or water conservation in a named city) ;			
	SPaG (3 marks) – award separately: **High performance (3)**: Learners spell and punctuate with consistent accuracy and consistently use rules of grammar; they use a wide range of specialist terms adeptly and with precision. **Intermediate performance (2)**: Learners spell and punctuate with considerable accuracy and use rules of grammar with general control; they use a good range of specialist terms with facility. **Threshold performance (1)**: Learners spell and punctuate with reasonable accuracy and use rules of grammar with some control; they use a limited range of specialist terms appropriately. **0 marks**: No marks awarded: no response or the spelling, punctuation and grammar is too weak, or the terminology is not relevant.			
03.1	40 (%)	Allow 38–42.	1	AO3 3.2.3.6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	coal fell from about 66% to about 2% ; renewables rose from about 2% to about 40% ; gas increased from about 2% to about 40% ; nuclear fell slightly (20% → 17%) ;	Any two with data.	2	AO3 3.2.3.6
03.3	government policy / targets to reduce CO_2 emissions (e.g. phasing out coal, Climate Change Act) ; coal is high carbon, contributes to global warming, and became uneconomic ; falling cost of wind and solar power ; gas power stations are cleaner than coal and flexible ; public concern about the environment / pressure for sustainability ;		4	AO2 3.2.3.6
03.4	USA is wealthier – people own more cars, appliances and larger homes ; India has many people still without electricity, and low incomes ; USA has more industry and more travel / long distances ; heating / air conditioning is used more in the USA ;		2	AO2/AO3 3.2.3.6
03.5	fossil fuel: burning releases CO_2, adding to the enhanced greenhouse effect / climate change ; air pollution (sulphur dioxide, acid rain) / mining damages landscapes / oil spills ; renewable: wind turbines are visually intrusive / kill birds ; HEP dams flood valleys ; solar farms use large areas of land ; development ;		4	AO2 3.2.3.7
03.6	A: reservoir (artificial lake) ; B: dam ; water is released from the reservoir through pipes (penstocks) ; the moving water turns turbines, which drive a generator ; the electricity is carried by power lines ;	C = power station (turbines / generator), D = power lines / pylons.	4	AO1/AO2 3.2.3.7
03.7	use energy-saving light bulbs / LED, switch off appliances ; improve home insulation / double glazing, so less heat is lost ; use public transport, walk or cycle, so fewer fossil fuels burned ; install smart meters / use more efficient appliances ; recycle: making products from recycled materials uses less energy ;		4	AO1/AO2 3.2.3.8

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.8	**Level 3 (5–6 marks)**: Detailed, accurate and relevant knowledge and understanding of the topic. Clear, well-developed and logically structured explanation/analysis with specific links made between ideas (and a use of place detail or the resources where asked). **Level 2 (3–4 marks)**: Some relevant and accurate knowledge and understanding. Some explanation/analysis, but developed unevenly or with limited links. Some use of detail or the resources. **Level 1 (1–2 marks)**: Basic, general knowledge with limited accuracy or relevance. Simple statements; little or no explanation or use of detail. **0 marks**: No relevant content.		6	AO1/AO2 3.2.3.8
	Indicative content (credit any other valid geography, and place-specific detail): • conservation reduces demand / saving money / less CO_2 released ; • insulation, efficient appliances, public transport ; • renewables (wind, solar, tidal, HEP, biomass) produce little CO_2 and will not run out ; • reduces dependence on imported fuel / improves energy security ; • limits: renewables can be intermittent (wind, sun) and costly to install ; conservation requires behaviour change ; • evaluation: combine both ; 'on balance...' ;			

Where marks are lost in Topic 6

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Supply and demand:** for each resource be clear about supply (where it comes from) and demand (who uses it, how much). 'Surplus' = supply > demand; 'deficit' = demand > supply.
- ☐ **Calculations:** show subtraction (available − used). Use units: m^3/person/year, calories/day, %, tonnes of oil equivalent.
- ☐ **Case-study detail:** for your chosen resource (food, water or energy) learn a named scheme with numbers: e.g. Lesotho Highlands (dams, 200 km tunnels, South Africa/Gauteng); Almería greenhouses; UK energy mix.
- ☐ **Positives and negatives:** for 6-mark 'evaluate/assess' questions give both. Add a judgement.
- ☐ **Sustainable:** mention that future generations need the resource. 'Sustainable' = meets needs now without harming the future.
- ☐ **Renewable vs non-renewable:** don't say renewables are 'free' or 'pollution-free'. Give a specific environmental impact.
- ☐ **Strategies:** distinguish conservation (using less) from increasing supply.
- ☐ **9-mark answers:** structure = for / against / judgement, with two named examples.